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Economic Studies

The link between energy prices and stock markets in European Union countries

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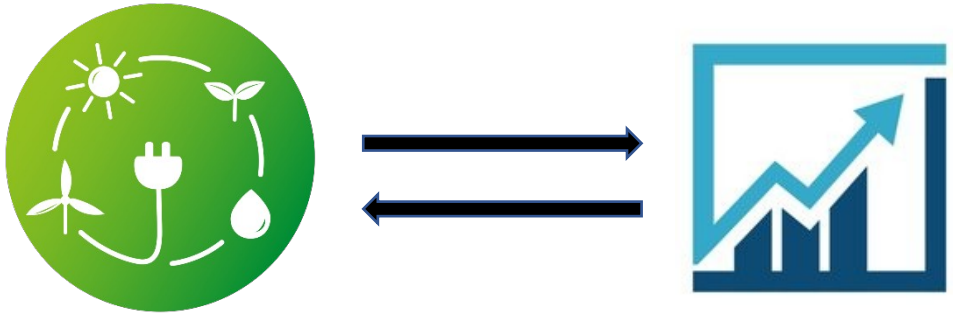
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Bucharest

Introduction

- A topic that has gained more importance in recent times is related to the impact generated by energy price volatility.
- Even though energy price dynamics can affect a wide spectrum of sectors and markets in different ways, one of the most sensitive to this type of movement is the capital market.
- This article analyzes the impact generated by energy prices, such as those of oil or gas, on the dynamics of the most important indices of EU countries.



Literature review

- Papers that analyze the subject of the link between energy prices and capital market returns for the US and other developed economies are:

  - ❑ Abhyankar et al. (2013) – [Structural VAR; for Japan]
  - ❑ Angelidis et al. (2015) – [Structural VAR, Markov Regime Switching, Probit regression model; for US]
  - ❑ Apergis and Miller (2009) – [Structural VAR; for Australia, Canada, France, Germany, Italy, Japan, the United Kingdom, and the United States]
  - ❑ Broadstock and Filis (2014) – [Structural VAR, Scalar-BEKK model; for China and US]
  - ❑ Chang et al. (2013) – [VARMA-GARCH, VARMA-AGARCH, DCC; for UK, US]
  - ❑ Kang et al. (2015a) – [Structural VAR; for US]
  - ❑ Kang et al. (2015b) – [Time-varying parameter structural VAR model; for US]

Methodology – data (I)

- For each of the EU countries, we consider in the analysis the main index of the stock market.
- For the energy prices, we used the generic future Brent Oil Price*, and for the gas, we selected the generic future European Natural Gas Price* (Dutch TTF).
- Data on indices, oil prices, and gas prices were taken from Bloomberg databases.
- The daily and monthly log-returns for each index were computed. The same algorithm was also applied in the case of oil and gas prices.
- We used daily data for the DCC-GARCH and Markov-Switching model (period 2010–2026) and monthly data for the S-VAR model (period 2010–2026). For the DCC-GARCH and MS-ARMA estimations, it was used OxMetrics, and for the S-VAR, it was used Matlab.

*Generic futures are created by Bloomberg, so you can analyze the historic performance of a futures contract. In order to analyze the historical performance of a future, generic futures link together the contracts that were in a certain position in the futures chain at a specific time.

Methodology – data (II)

- To create the regional indices, we used the EU regions according to EuroVoc (Figure 1). To group the countries for the indices based on industrial production development, we used Eurostat data (Figure 2).

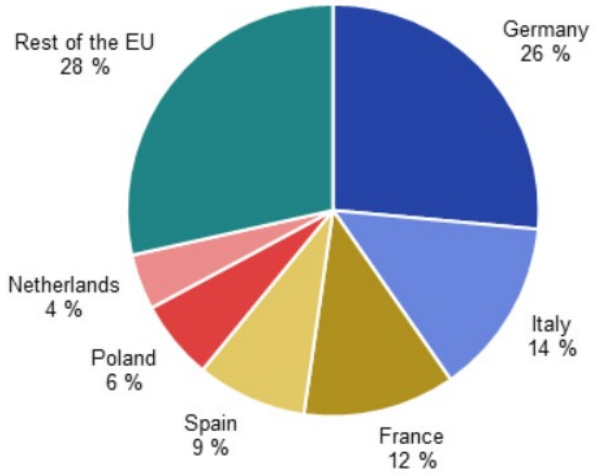
- The weight associated with each country in the indices was calculated based on the stock market capitalization.

Figure 1. Regions of Europe



Source: EuroVoc

Figure 2. EU value of sold industrial production, by country, 2024 (% of total value of sold production)



Source: Eurostat

Methodology – models (I)

- The first model used in the analysis was the Dynamic Conditional Correlation GARCH (DCC-GARCH), developed by Engle and Sheppard (2001) as an extension of the CCC-GARCH. In this case, the conditional covariance matrix is computed as follows:

$$\mathbf{H}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$$

- Where H_t is the conditional covariance matrix, D_t is the diagonal matrix containing standard deviations of the fitted univariate GARCH models for each asset, and the R_t is the correlation matrix. Considering that D_t has the form:

$$\mathbf{D}_t = \begin{pmatrix} h_{1,t}^{1/2} & 0 & \cdots & 0 \\ 0 & h_{2,t}^{1/2} & \ddots & \vdots \\ 0 & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & h_{n,t}^{1/2} \end{pmatrix}$$

- The calculus for H_t becomes:

$$[\mathbf{H}_t]_{ij} = h_{it}^{1/2} h_{jt}^{1/2} \rho_{ij,t}$$

- The idea of the models in this class is that the covariance matrix (H_t) can be decomposed into conditional standard deviations (D_t) and a correlation matrix (R_t). In the DCC-GARCH model, both conditional standard deviations and the correlation matrix are designed to be time-varying.

- This type of model is used in predicting the dependence in the comovements of asset returns.

Methodology – models (II)

- Another model used in the analysis was the Markov-Switching ARMA model (MS-ARMA). This approach allows the analysis to capture structural changes and nonlinear dynamics in the relationship between oil/gas prices and financial markets.

- The model specification:

$$y_t = \mu_{s_t} + \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \varepsilon_t$$

- Where s represents the unobserved market regime. Different regimes reflect varying levels of correlation persistence and market dependence.

- The regime transition process:

$$P_{ij} = P(s_t = j \mid s_{t-1} = i)$$

- This measures the probability of switching between high- and low-correlation regimes.

Results (I) – EU regions and oil price

Figure 3. Conditional correlation of the CEE region index with oil price

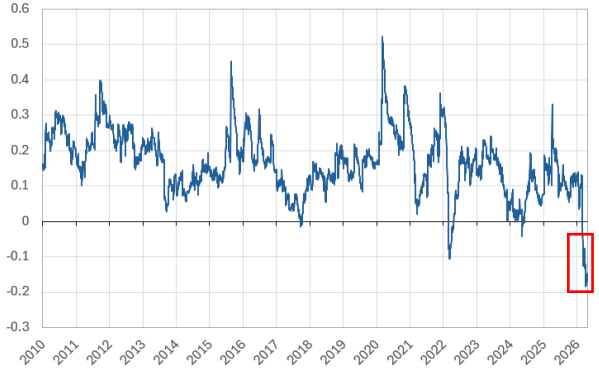


Figure 4. Conditional correlation of the NE region index with oil price



Figure 5. Conditional correlation of the SE region index with oil price

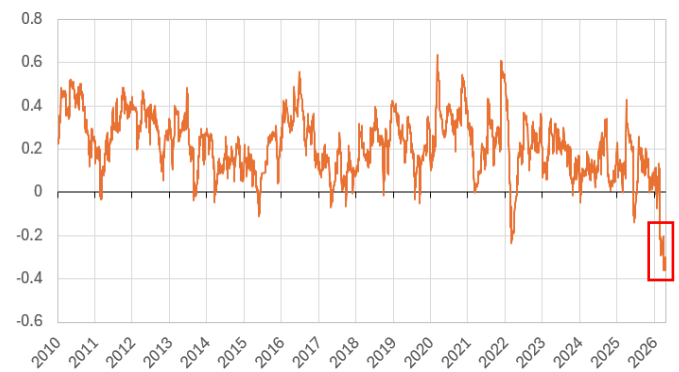
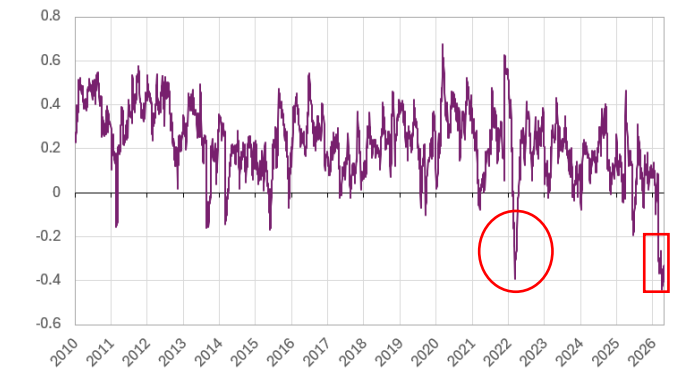


Figure 6. Conditional correlation of the WE region index with oil price



Source: author's calculations

- For all EU regions, the conditional correlation of the stock market index with oil price dynamics is generally positive, in most cases in the range of 0 – 0.4.
- In the first part of 2022, a period associated with the Russian invasion of Ukraine, the negative correlation between oil prices and stock indices was felt with greater intensity in the Western Europe region.
- Geopolitical tensions in the Middle East contributed to the increase in the negative correlation between oil price dynamics and European stock market indices in 2026.

Results (II) – EU regions and oil price

Figure 7. The regime with high conditional correlation of the CEE region index with oil price

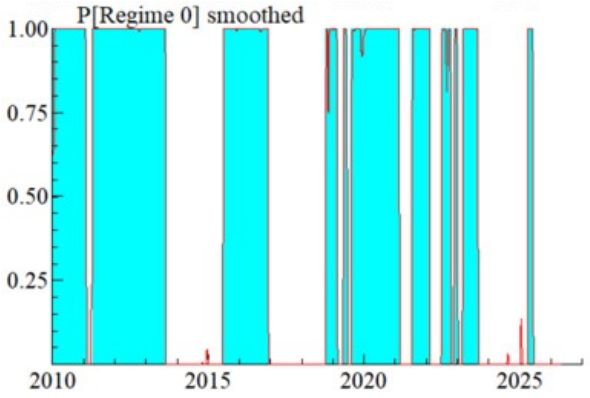


Figure 8. The regime with high conditional correlation of the NE region index with oil price

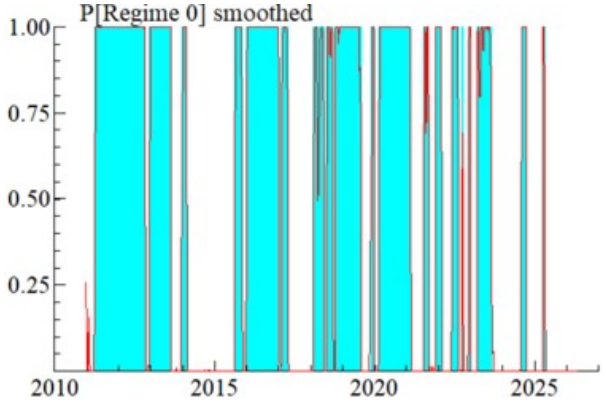


Figure 9. The regime with high conditional correlation of the SE region index with oil price

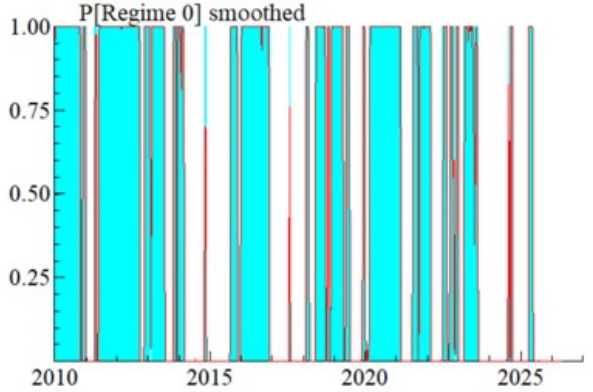


Figure 10. The regime with high conditional correlation of the WE region index with oil price

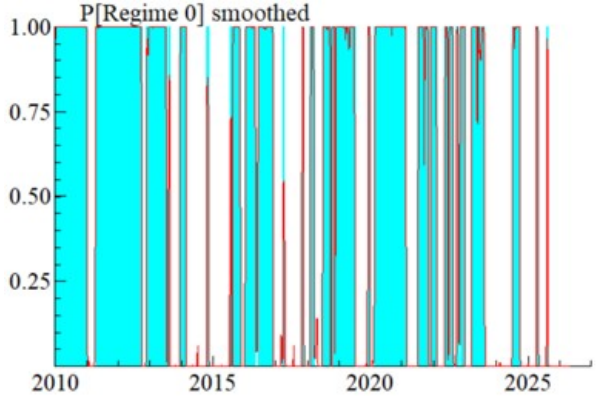


Figure 11. The regime with low conditional correlation of the CEE region index with oil price

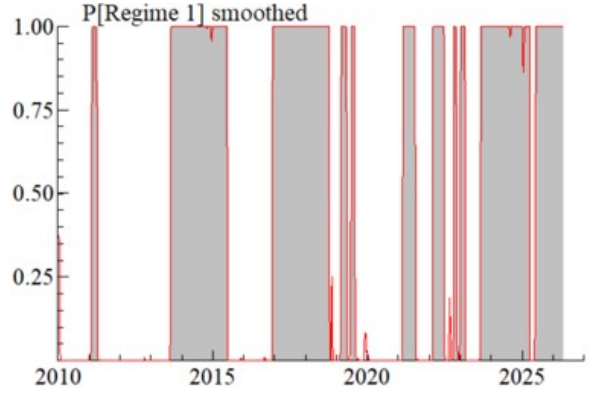


Figure 12. The regime with low conditional correlation of the NE region index with oil price

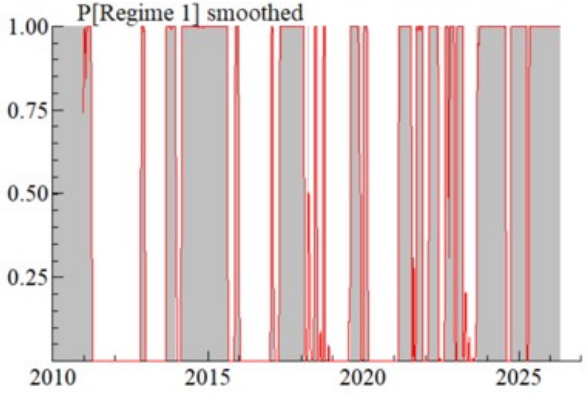


Figure 13. The regime with low conditional correlation of the SE region index with oil price

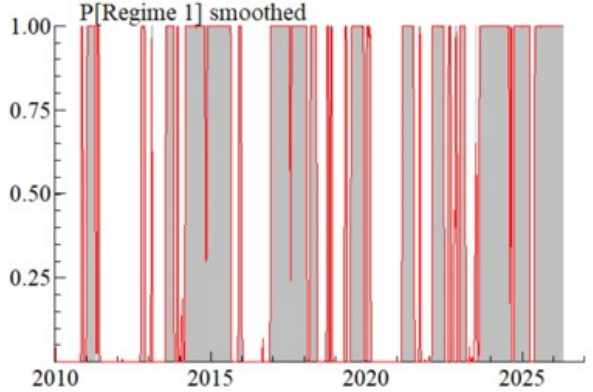
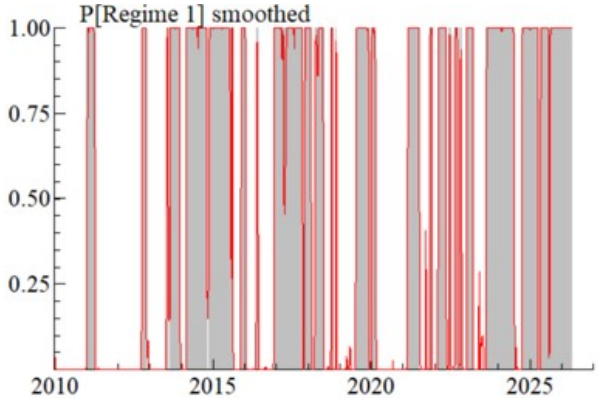


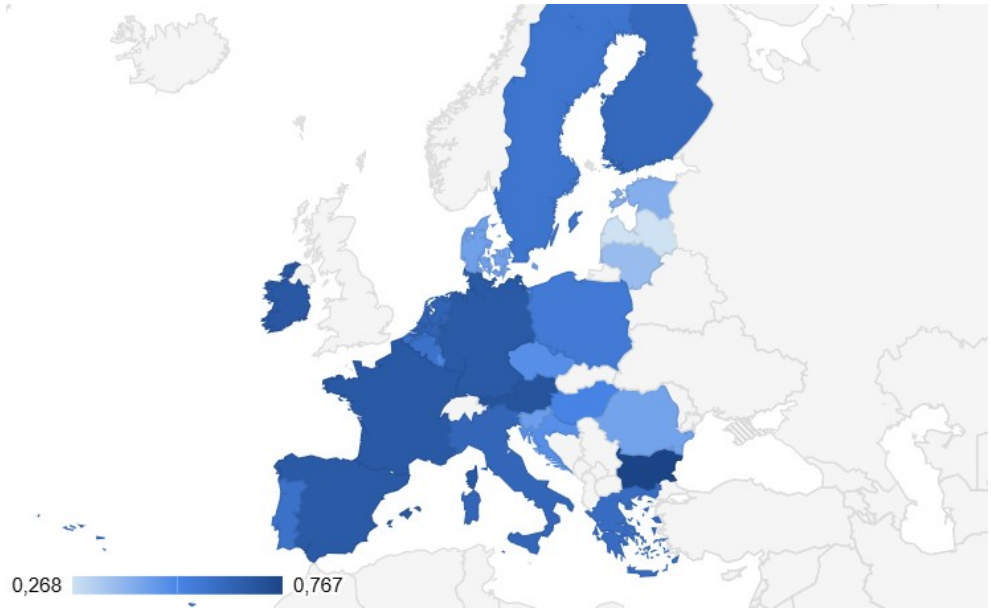
Figure 14. The regime with low conditional correlation of the WE region index with oil price



Results (III) – SE and WE most correlated with oil price

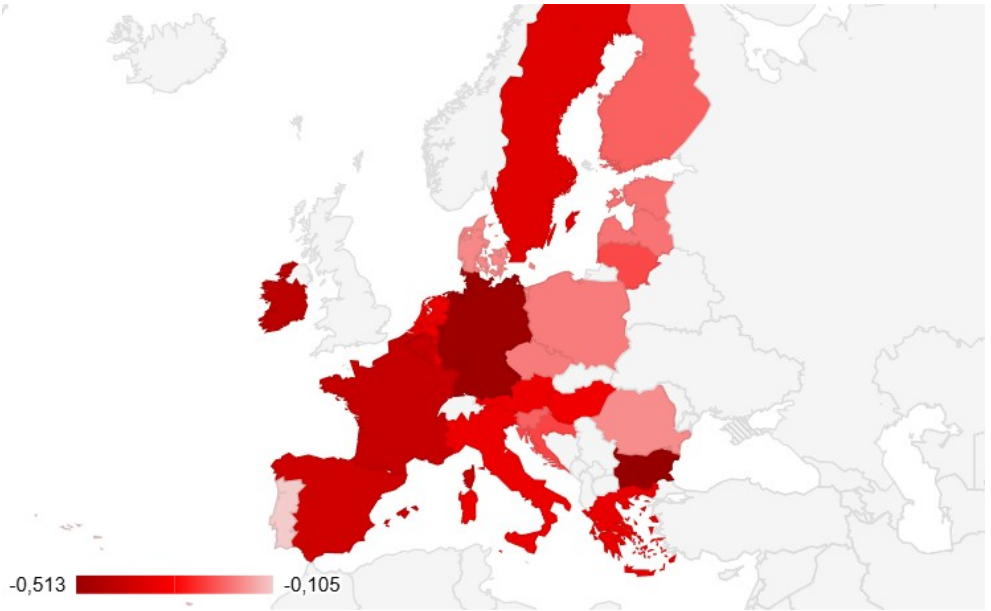
- In general, there was a stronger correlation between the oil price and the main indices of the countries in the western and southern parts of Europe.

Figure 15. The highest level of conditional correlation of the country index with oil price registered during the studied period



Source: author's calculations

Figure 16. The lowest level of conditional correlation of the country index with oil price registered during the studied period

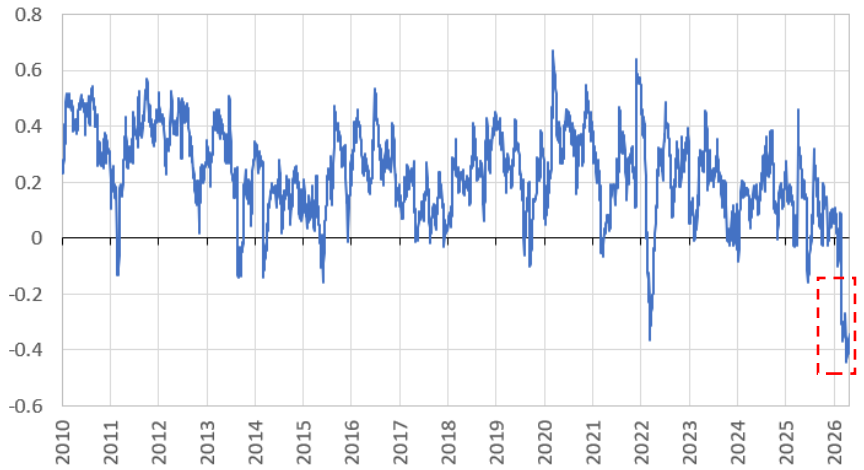


Source: author's calculations

Results (IV) – Industrialized countries and oil price

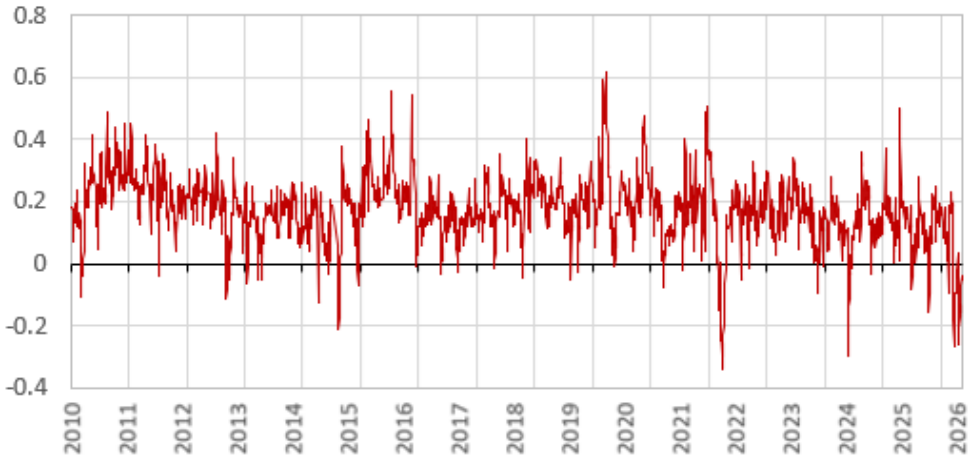
- During the studied period, the price of oil was more strongly correlated with the stock market index of the countries with high industrial production compared to the rest of the EU states.
- As a result of the Russian invasion of Ukraine, the correlation between the price of oil and the stock market indices was more negative in the case of highly industrialized states.
- The Middle East tensions intensified the negative relationship between oil prices and European equity indices, particularly in economies with more developed industrial sectors.

Figure 17. Conditional correlation of the index for most industrialized EU countries with oil price



Source: author's calculations

Figure 18. Conditional correlation of the index for less industrialized EU countries with oil price



Source: author's calculations

Results (V) - EU regions and gas price

Figure 19. Conditional correlation of the CEE region index with gas price

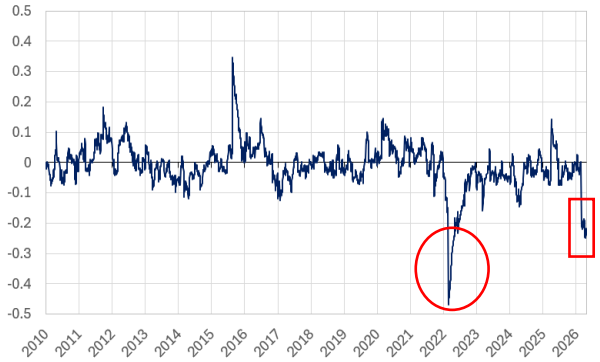


Figure 20. Conditional correlation of the NE region index with gas price



Figure 21. Conditional correlation of the SE region index with gas price

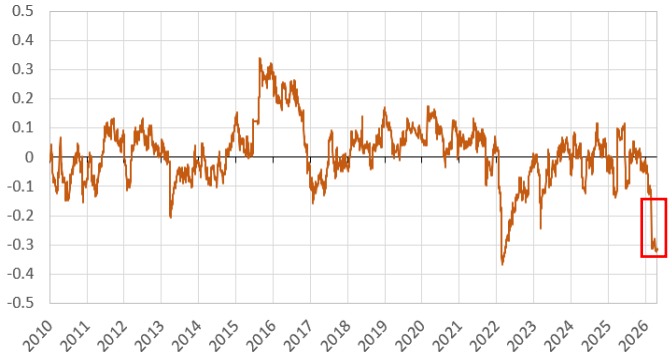


Figure 22. Conditional correlation of the WE region index with gas price



Source: author's calculations

- For most European regions, the conditional correlation of stock indices with gas price dynamics is generally lower than that recorded with respect to oil prices.
- Following the Russian invasion of Ukraine in early 2022, the CEE region experienced the strongest negative correlation between gas prices and stock indices. Another negative correlation peak was observed after the annexation of Crimea in March 2014, particularly in the NE and WE regions.
- Geopolitical tensions in the Middle East contributed to the increase in the negative correlation between gas price dynamics and European stock market indices in 2026.

Results (VI) – EU regions and gas price

Figure 23. The regime with high conditional correlation of the CEE region index with gas price

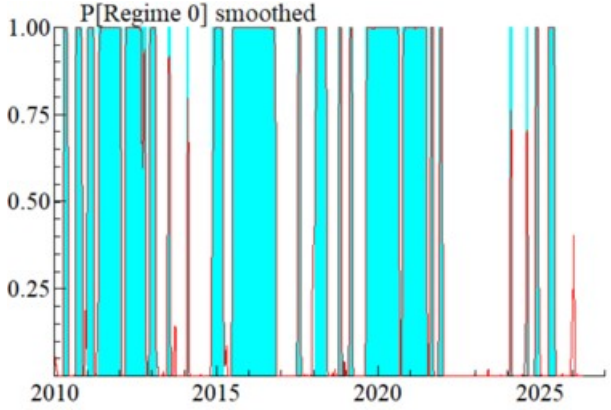


Figure 24. The regime with high conditional correlation of the NE region index with gas price

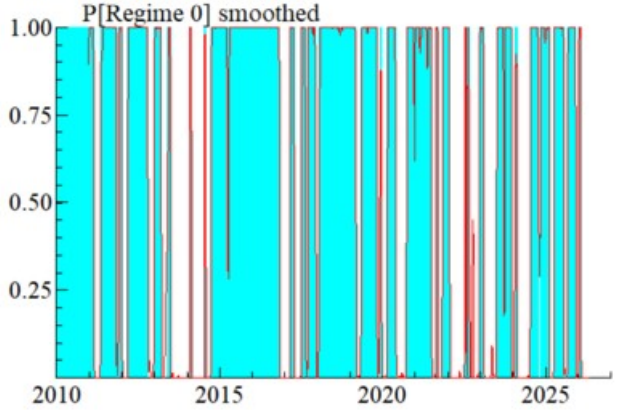


Figure 25. The regime with high conditional correlation of the SE region index with gas price

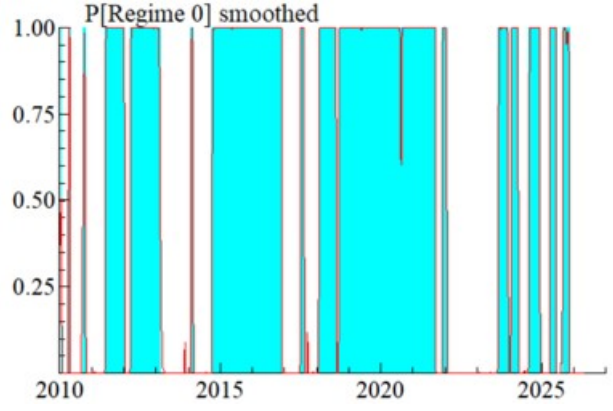


Figure 26. The regime with high conditional correlation of the WE region index with gas price

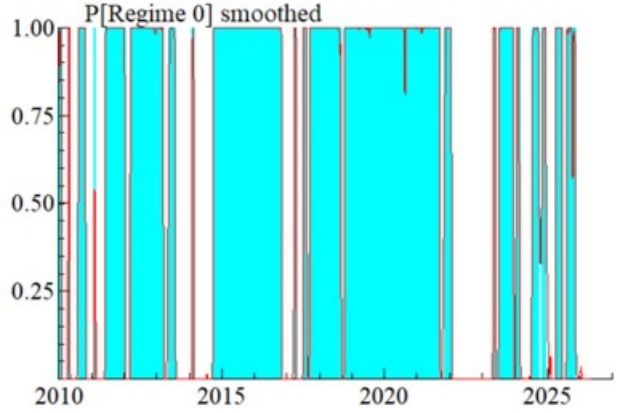


Figure 27. The regime with low conditional correlation of the CEE region index with gas price

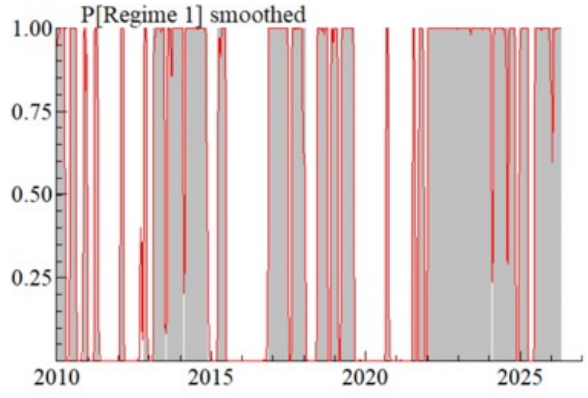


Figure 28. The regime with low conditional correlation of the NE region index with gas price

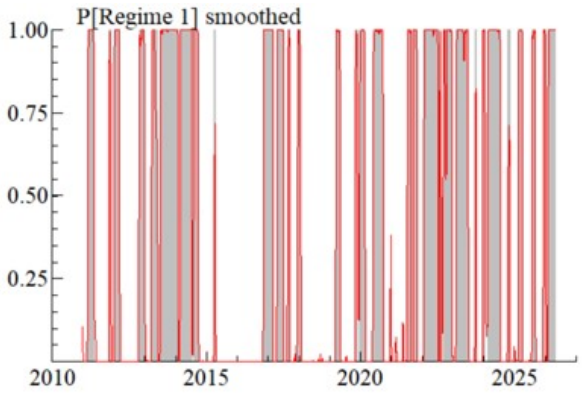


Figure 29. The regime with low conditional correlation of the SE region index with gas price

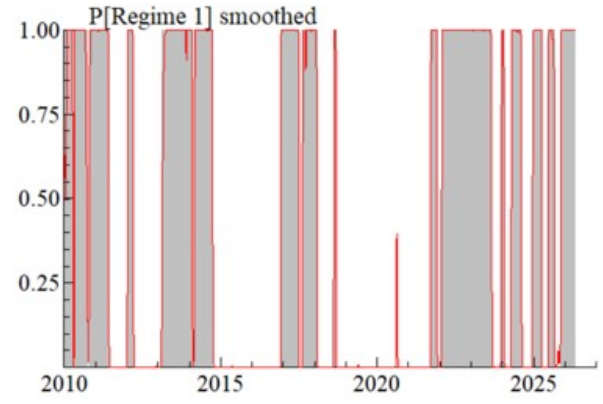
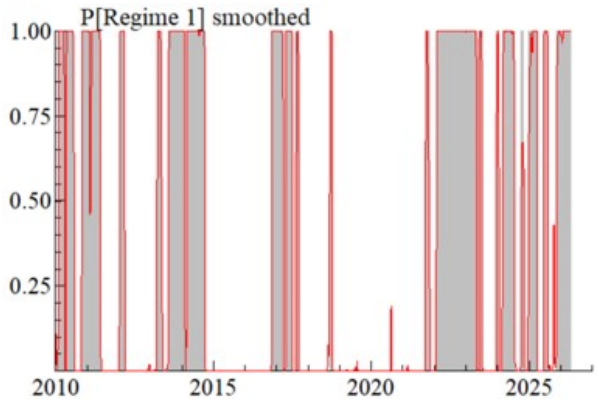


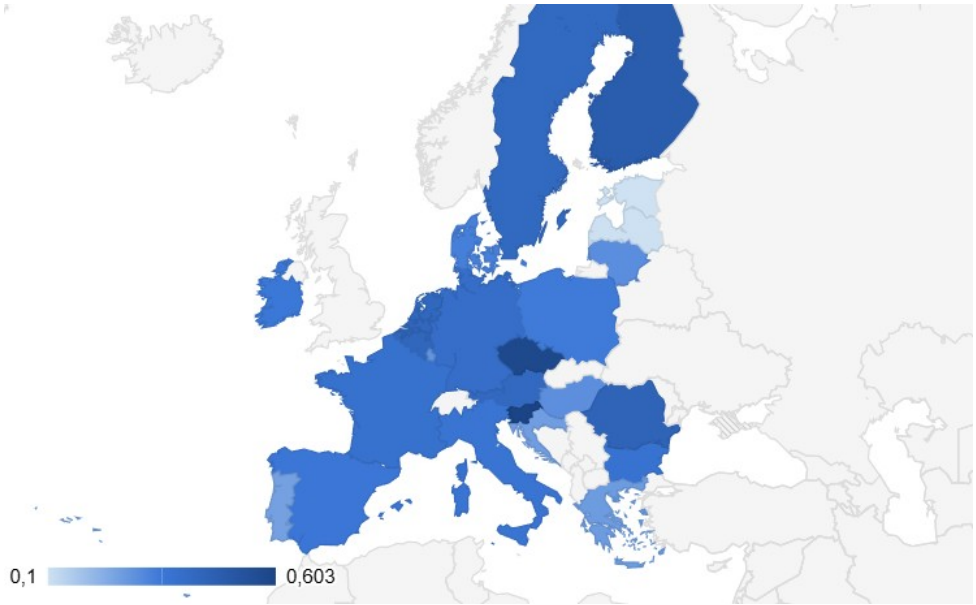
Figure 30. The regime with low conditional correlation of the WE region index with gas price



Results (VII) – NE&WE most correlated with gas price

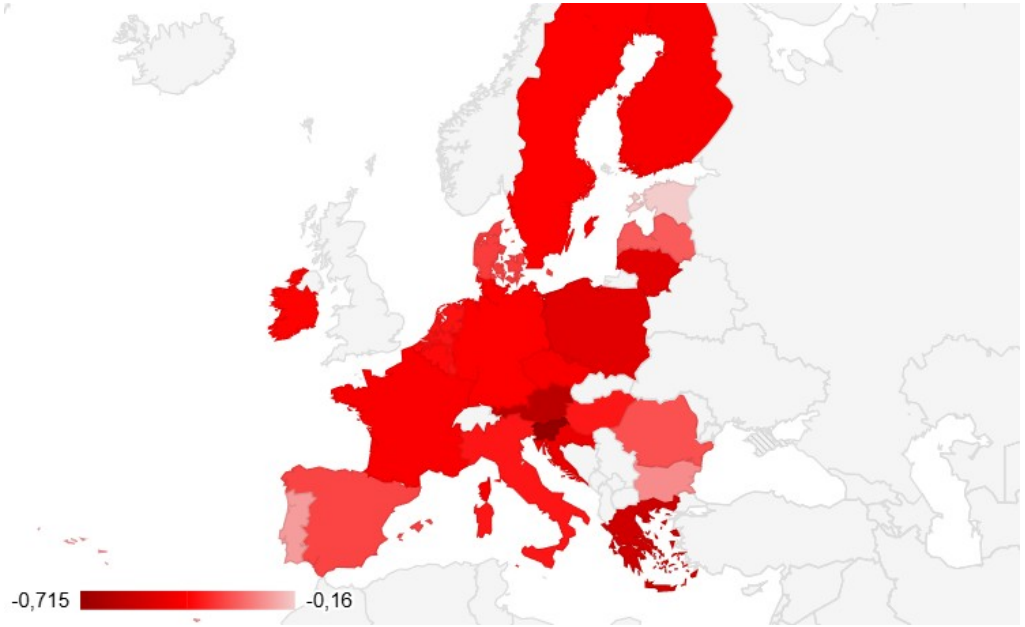
- In general, there was a stronger correlation between the gas price dynamics and the main indices of the countries in the northern and western part of Europe.

Figure 31. The highest level of conditional correlation of the country index with gas price, during the studied period



Source: author's calculations

Figure 32. The lowest level of conditional correlation of the country index with gas price, during the studied period

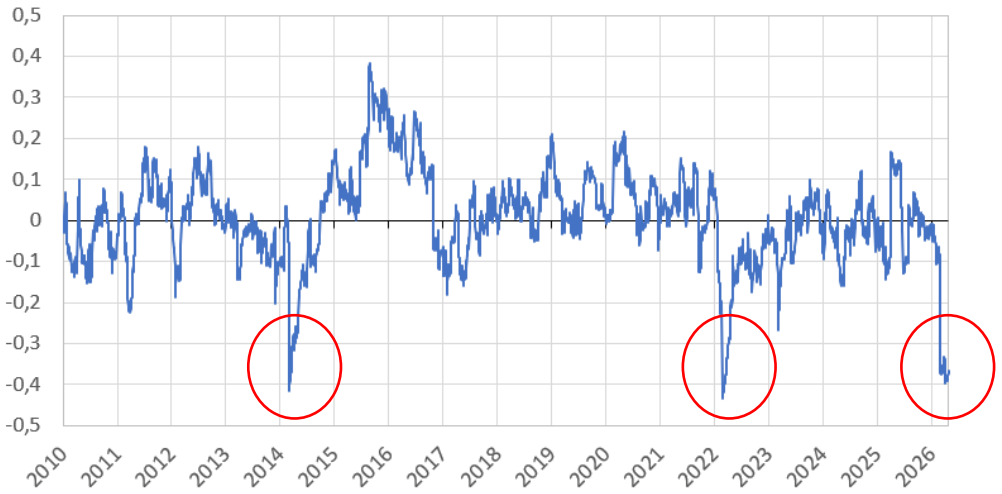


Source: author's calculations

Results (VIII) - Industrialized countries and gas price

- During periods of geopolitical tension, such as the annexation of Crimea, the beginning of the war in Ukraine, or the Middle East tensions in 2026, the most industrialized EU countries exhibited a stronger negative correlation between gas price and stock market indices compared to the rest of the states.

Figure 33. Conditional correlation of the index for most industrialized EU countries with gas price



Source: author's calculations

Figure 34. Conditional correlation of the index for less industrialized EU countries with gas price



Source: author's calculations

Results (IX) – EU regions response to oil price shock

Figure 35. Response of CEE region index to the oil price shock

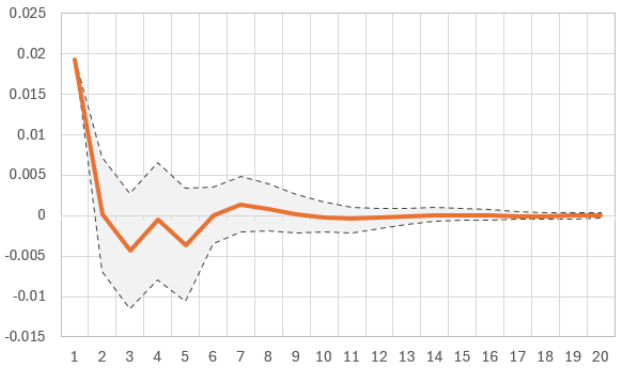


Figure 36. Response of NE region index to the oil price shock

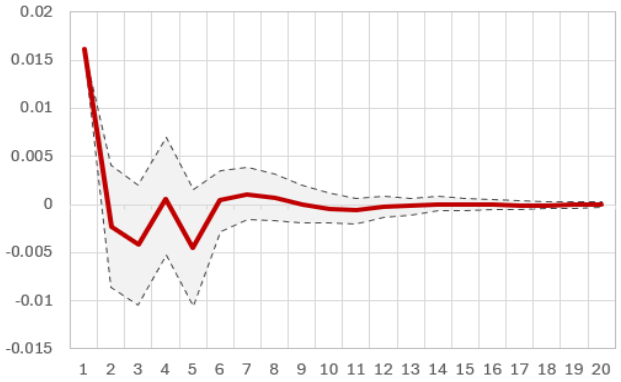


Figure 37. Response of SE region index to the oil price shock

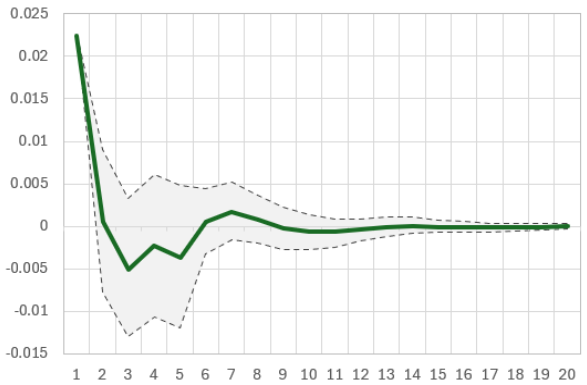
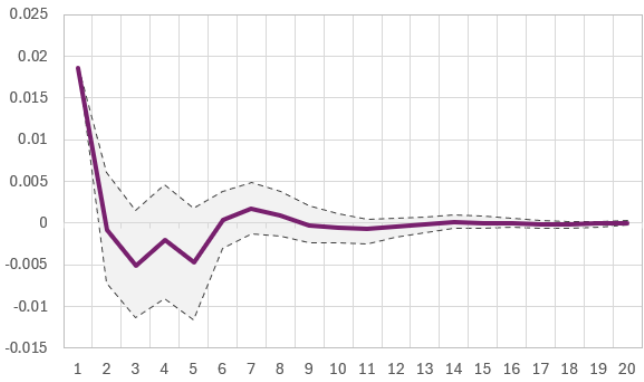


Figure 38. Response of WE region index to the oil price shock



Note: the dotted lines represent the 95% confidence interval limits

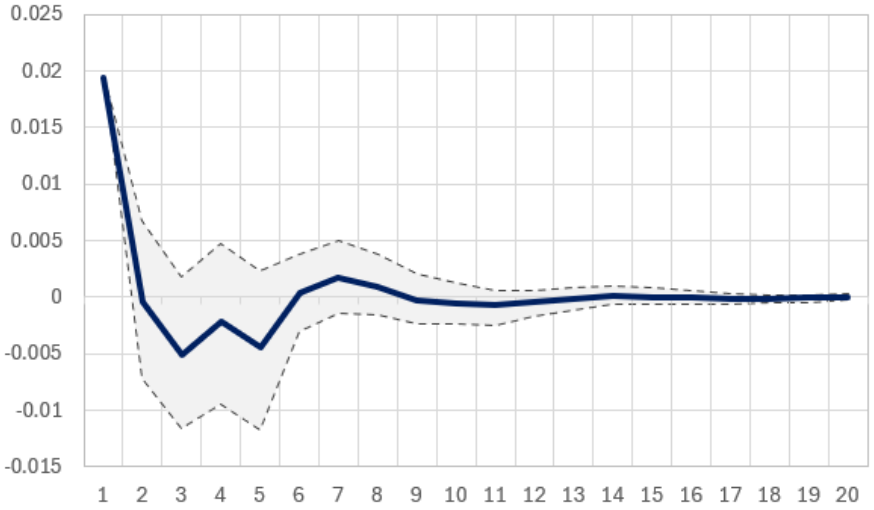
Source: author's calculations

- The impact generated by a shock at the level of oil price does not differ significantly from one EU geographical region to another. The impact is most pronounced in the months immediately following the shock, gradually fading over the medium and long term.

Results (X) - Industrialized countries and oil price shock

- In the short term (first few months after the occurrence of the oil price shock), the stock indices of the most industrialized EU countries are slightly more affected compared to the rest of the EU states.

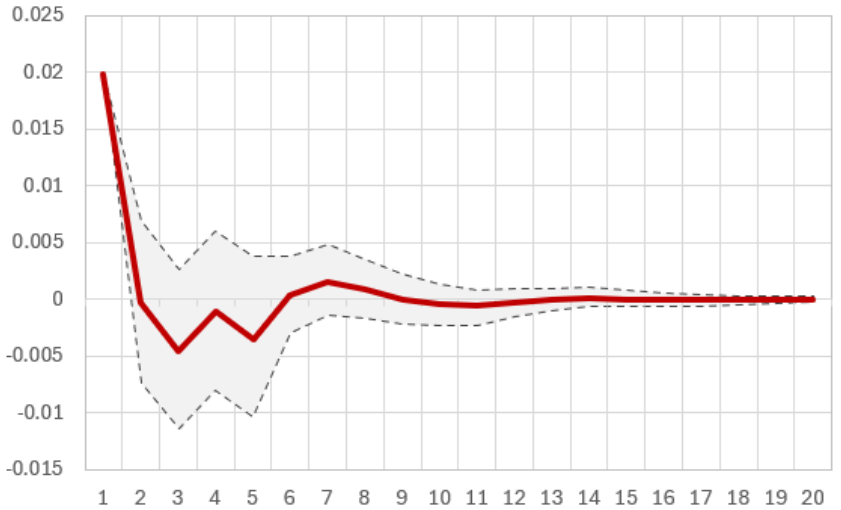
Figure 39. Response of the index for the most industrialized EU countries to the oil price shock



Note: the dotted lines represent the 95% confidence interval limits

Source: author's calculations

Figure 40. Response of the index for the less industrialized EU countries to the oil price shock



Note: the dotted lines represent the 95% confidence interval limits

Source: author's calculations

Results (XI) - EU regions response to gas price shock

Figure 41. Response of CEE region index to the gas price shock

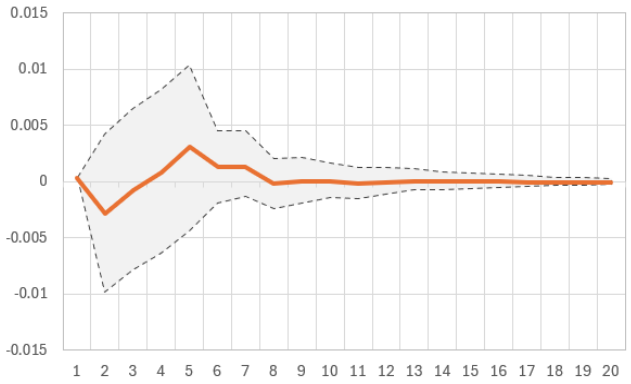


Figure 42. Response of NE region index to the gas price shock

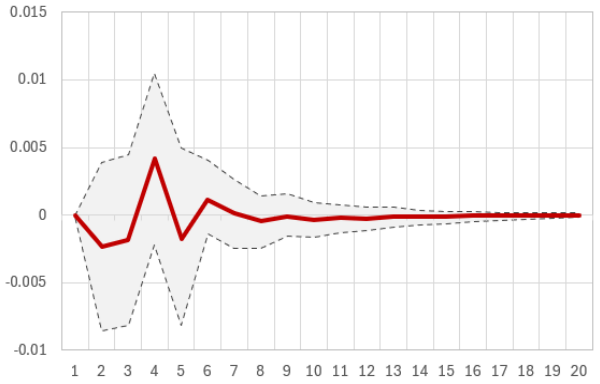


Figure 43. Response of SE region index to the gas price shock

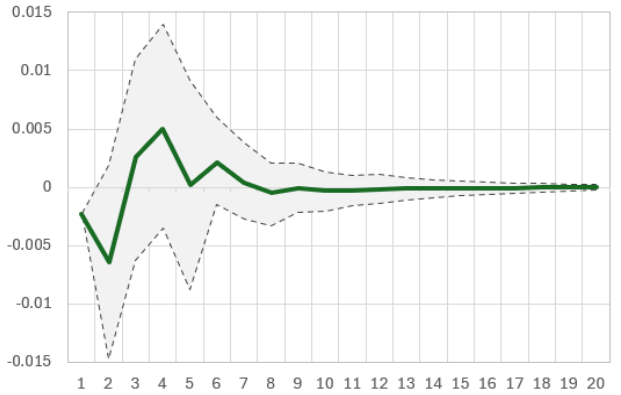
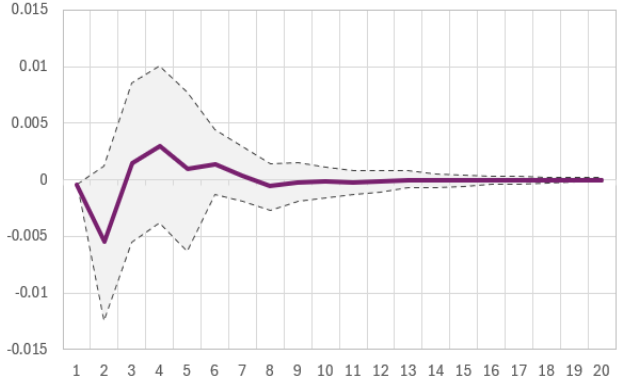


Figure 44. Response of WE region index to the gas price shock



Note: the dotted lines represent the 95% confidence interval limits

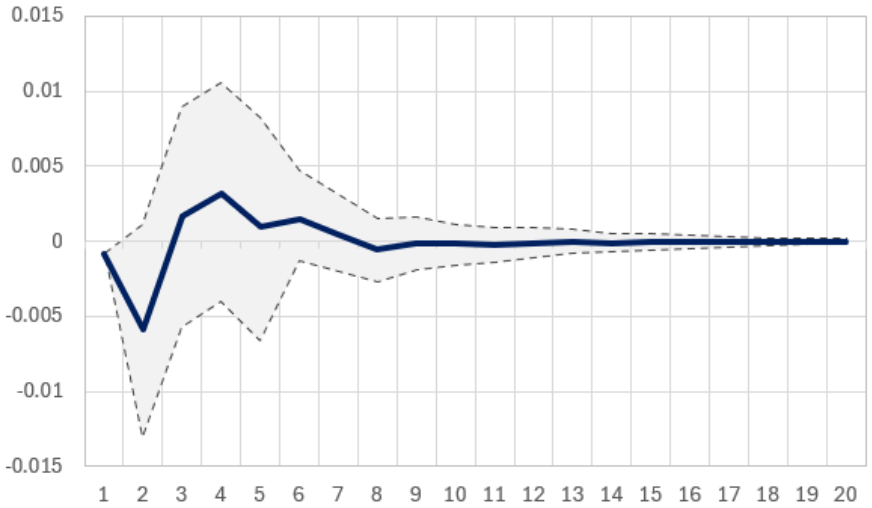
Source: author's calculations

- On the short term (first few months after the gas price shock occurs) the stock indices of countries from South Europe and West Europe regions are more strongly affected, in comparison with the ones from Central and Eastern Europe or Northern Europe region.

Results (XII) - Industrialized countries and gas price shock

- In the short term (the first few months after the occurrence of the gas price shock), the stock indices of the most industrialized countries in the EU are more strongly affected in comparison with the rest of the EU states.

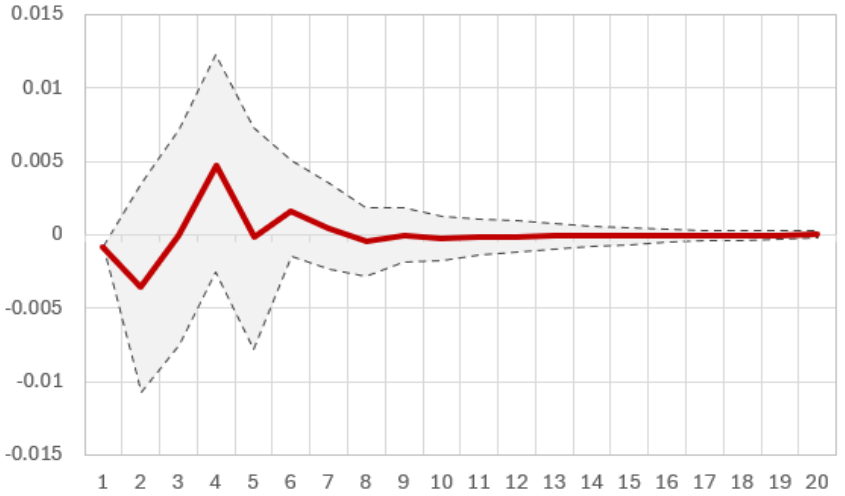
Figure 45. Response of the index for the most industrialized EU countries to gas price shock



Note: the dotted lines represent the 95% confidence interval limits

Source: author's calculations

Figure 46. Response of the index for the less industrialized EU countries to gas price shock



Note: the dotted lines represents the 95% confidence interval limits

Source: author's calculations

Conclusions

1. For most EU regions, the conditional correlation of stock indices with gas price dynamics is generally lower than that recorded with respect to oil prices.

2. Stock market indices in the SE and WE are generally strongly correlated with oil price dynamics compared to indices in the rest of the EU regions.

3. The price of oil is strongly correlated with the stock market indices of the countries with high industrial production, compared to the rest of the EU states.

4. Geopolitical tensions in the Middle East contributed to the increase in the negative correlation between oil/gas price dynamics and European stock market indices in 2026.

5. Stock market indices in the NE and WE are generally strongly correlated with gas price dynamics compared to indices in the rest of the EU regions.

6. As a result of the Russian invasion of Ukraine, the negative correlation between the gas price and stock indices was felt with greater intensity in the CEE region.

7. In the short term, the stock indices of the EU industrialized countries are strongly affected by the effects of the oil and gas price shocks.

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